ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 5 - Power-Supply Filters & Voltage Regulators

14 July 2026

Today's Agenda — Lecture Notes pp. 66–74

- **Part 1 — Power-Supply Filters**
  - The capacitor-input filter: how it charges, discharges, and produces ripple
  - Ripple factor, the filter-output formulas, surge current, and fusing
- **Part 2 — Voltage Regulators**
  - Why a regulator is added, and the three-terminal IC regulator
  - Line regulation and load regulation, with worked examples
- **Part 3 — Wrap-Up**
  - The complete power supply, key takeaways, and a look ahead to the Zener diode

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
  - Describe how a capacitor-input filter smooths a rectified voltage
  - Define ripple voltage and calculate the ripple factor
  - Find the ripple and dc output of a filtered full-wave rectifier
  - Explain surge current and how a slow-blow fuse protects the supply
  - Explain what a voltage regulator does and describe a three-terminal IC regulator
  - Calculate line regulation and load regulation

Part 1 — Power-Supply Filters

Where We Left Off — Rectifiers

- **Recap**
  - A rectifier turns AC into pulsating DC — but it still pulses (60 Hz half-wave, 120 Hz full-wave)
  - Electronic circuits need a steady, constant DC level

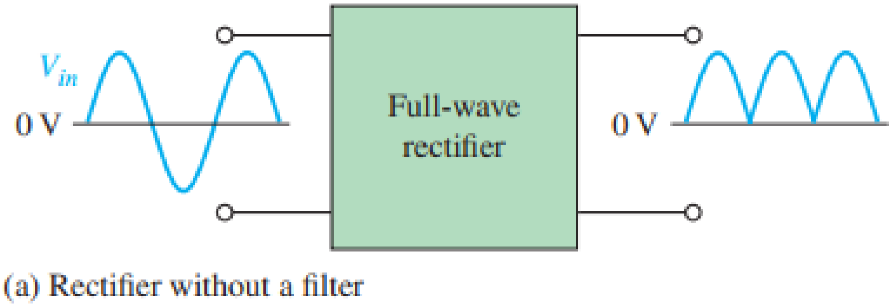


Fig 2.42 (a)— Power supply filtering: unfiltered rectifier output

Today's Focus — Filters

- **The Filter's Job**
  - A filter smooths the pulsating rectifier output into nearly constant DC
  - Filters are built from capacitors
  - The small leftover fluctuation is called ripple — less ripple is better

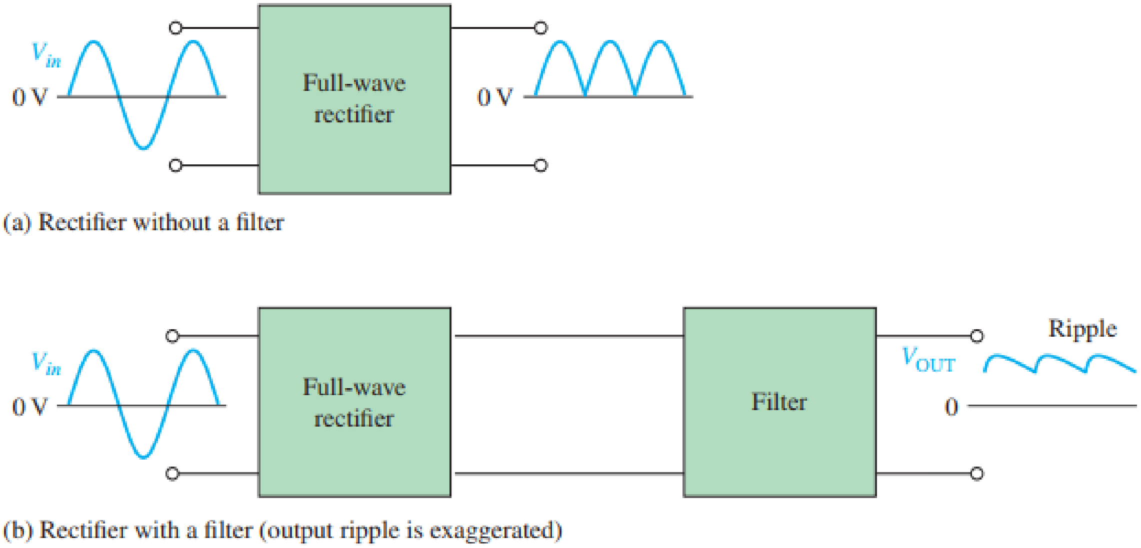


Fig 2.42 — Power supply filtering

The Capacitor-Input Filter — Charging

- **The Circuit**
  - A capacitor is connected from the rectifier output to ground; R_L is the load
- **Charging Up**
  - On the first positive quarter-cycle, the diode is forward-biased
  - The capacitor charges quickly to within 0.7 V of the input peak

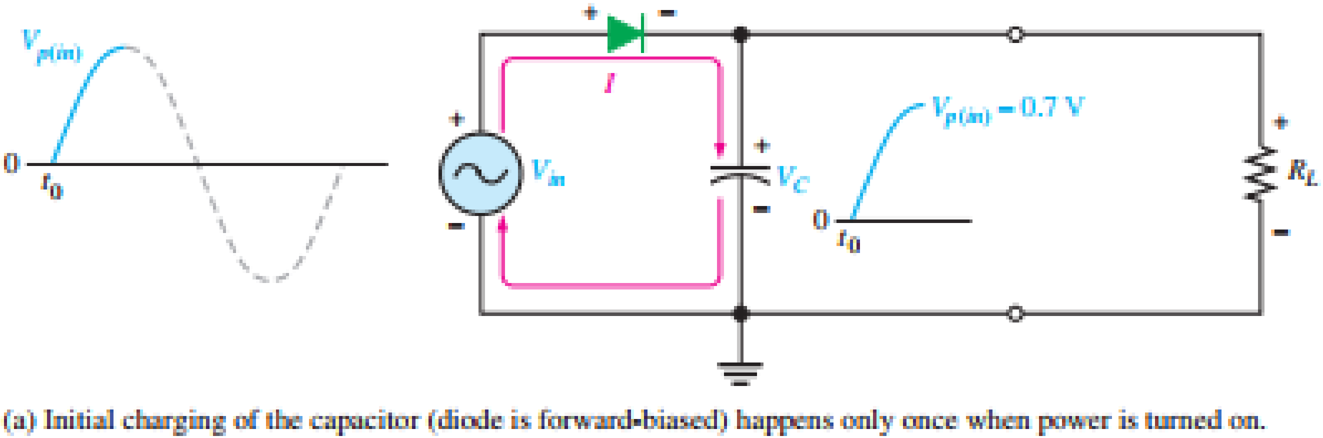
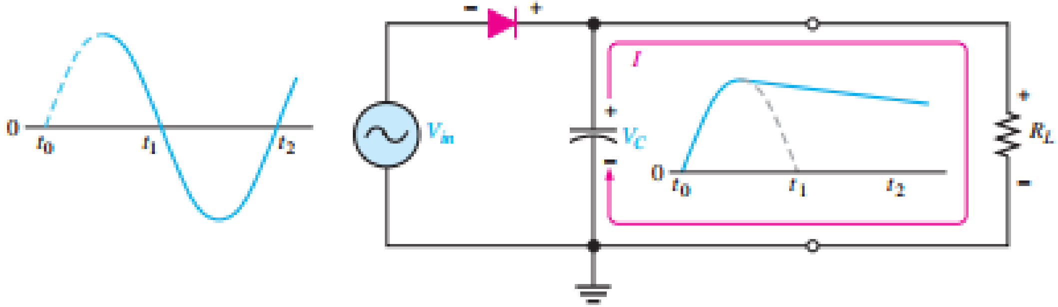
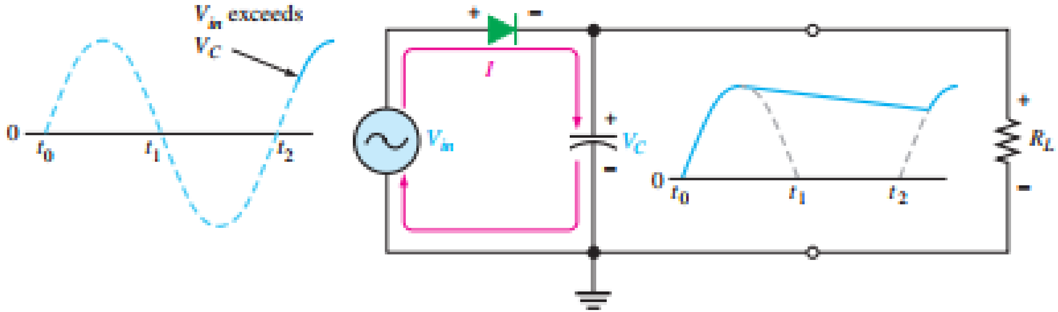


Fig 2.43 — Capacitor-input filter: the capacitor charges to the peak, holds, then recharges each cycle.

The Capacitor-Input Filter — Charging Cont'd



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.

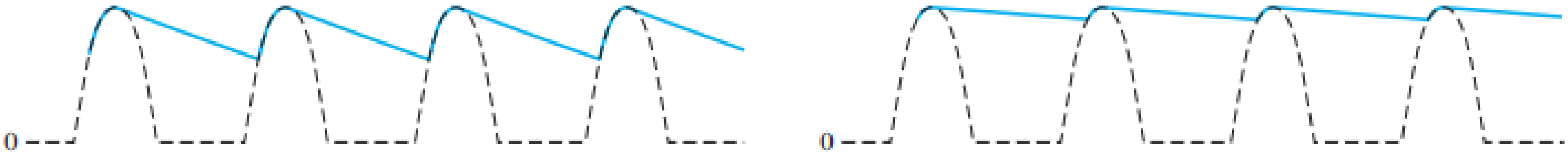


(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid dark blue curve.

Fig 2.43 — Capacitor-input filter: the capacitor charges to the peak, holds, then recharges each cycle.

Ripple Voltage

- **Holding the Charge**
  - As the input falls below the peak, the diode becomes reverse-biased
  - The capacitor can only discharge slowly through R_L — set by the $R_L \cdot C$ time constant
  - A long time constant ($R_L \cdot C$ much greater than the input period) means little discharge between peaks
- **The Result — Ripple**
  - Ripple voltage is the variation in capacitor voltage due to charging and discharging
  - The smaller the ripple, the better the filtering action



(a) Larger ripple (blue) means less effective filtering.

(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

Fig 2.44 — Half-wave ripple voltage

Full-Wave vs Half-Wave Frequency

- **The Key Difference: Output Frequency**
  - A half-wave rectifier's output pulses once per input cycle (60 Hz)
  - A full-wave rectifier's output pulses twice per cycle (120 Hz) — its period is HALF
- **Why This Helps Filtering**
  - Shorter time between peaks → the capacitor discharges less before the next peak tops it up
  - So for the same capacitor and load, the full-wave output has smaller ripple

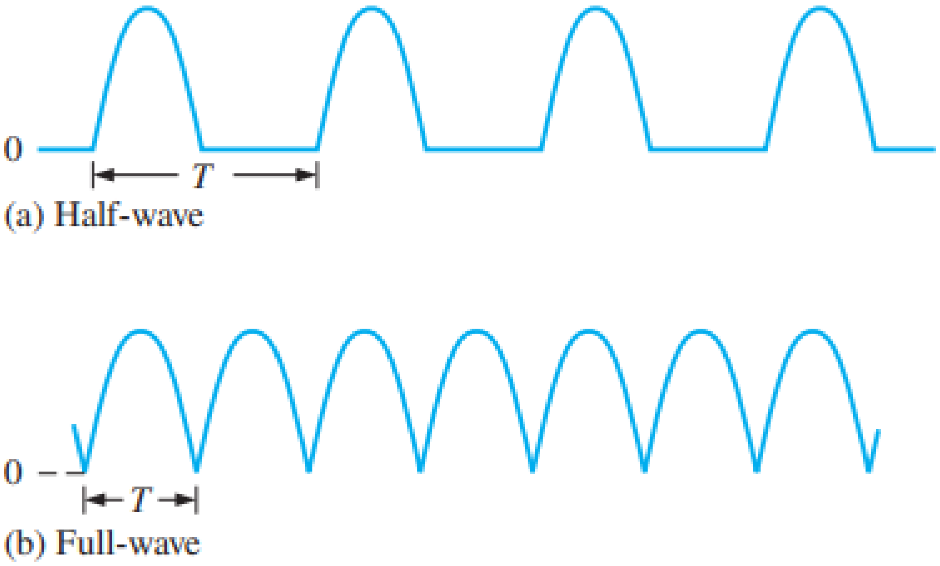


Fig 2.45 — A full-wave output has half the period — twice the frequency — of a half-wave output.

Half-Wave vs. Full-Wave Ripple

- **Full-Wave Filters Better**
  - A full-wave output pulses at 120 Hz — twice as often as half-wave (60 Hz)
  - Shorter time between peaks → the capacitor discharges less
- **The Payoff**
  - For the same C and RL, full-wave has smaller ripple than half-wave
  - This is one more reason full-wave rectifiers are preferred

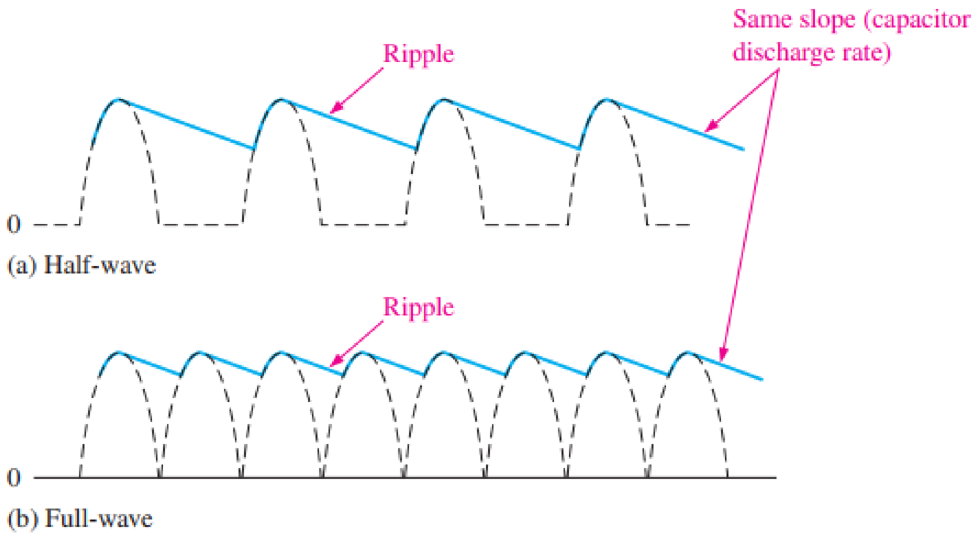


Fig 2.46 — Half-wave vs full-wave ripple for the same capacitor and load.

The Ripple Factor

- **A Measure of Filtering**
  - The ripple factor compares the ripple to the DC level:

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

  - Vr(pp) is the peak-to-peak ripple; V_{DC} is the DC (average) output

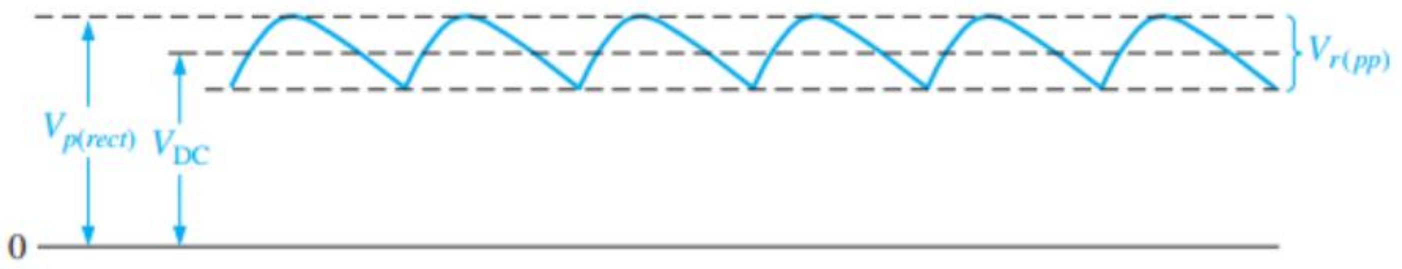


Fig 2.47 — Vr(pp) and VDC together define the ripple factor.

- **Lower Is Better**
  - A smaller r means smoother DC
  - Lower the ripple factor by increasing the filter capacitor C or the load resistance R_L

Filter-Output Formulas (Full-Wave)

- **For a Full-Wave Capacitor-Input Filter**
  - Peak-to-peak ripple:

$$V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)}$$

  - DC output:

$$V_{DC} \cong \left(1 - \frac{1}{2fR_LC}\right)V_{p(rect)}$$

- **Reading the Formulas**
  - Vp(rect) is the unfiltered peak rectified voltage; f is the ripple frequency (120 Hz for full-wave)
  - Increasing R_L or C lowers the ripple and raises the dc output

Worked Example — Ripple Factor

- **The Question (Example 2-8)**
  - Determine the ripple factor for the filtered bridge rectifier with a load as indicated in Figure 2-48.

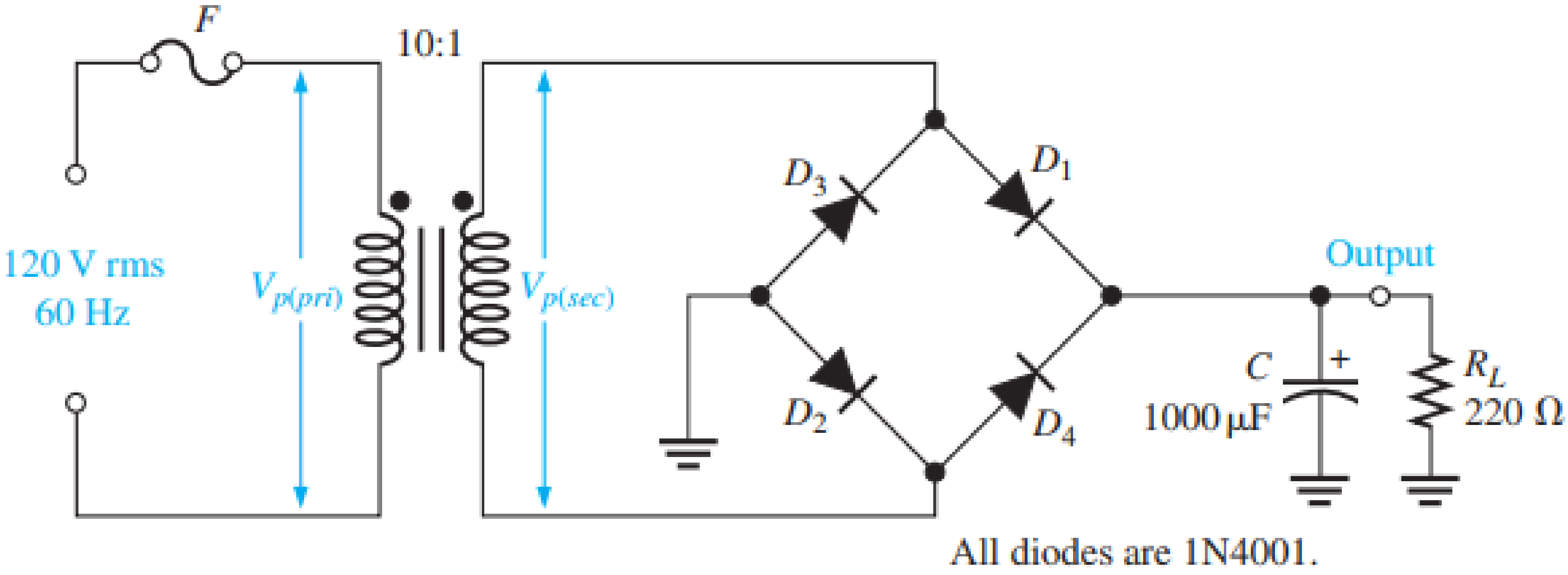


Fig 2.48 — Filtered bridge rectifier for Example 2–48.

Worked Example — Ripple Factor

The transformer turns ratio is $n = 0.1$. The peak primary voltage is

$$V_{p(pri)} = 1.414V_{rms} = 1.414(120\text{ V}) = 170\text{ V}$$

The peak secondary voltage is

$$V_{p(sec)} = nV_{p(pri)} = 0.1(170\text{ V}) = 17.0\text{ V}$$

The unfiltered peak full-wave rectified voltage is

$$V_{p(rect)} = V_{p(sec)} - 1.4\text{ V} = 17.0\text{ V} - 1.4\text{ V} = 15.6\text{ V}$$

The frequency of a full-wave rectified voltage is 120 Hz. The approximate peak-to-peak ripple voltage at the output is

$$V_{r(pp)} \cong \left(\frac{1}{fR_LC}\right)V_{p(rect)} = \left(\frac{1}{(120\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})}\right)15.6\text{ V} = 0.591\text{ V}$$

The approximate dc value of the output voltage is determined as follows:

$$V_{DC} = \left(1 - \frac{1}{2fR_LC}\right)V_{p(rect)} = \left(1 - \frac{1}{(240\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})}\right)15.6\text{ V} = 15.3\text{ V}$$

The resulting ripple factor is

$$r = \frac{V_{r(pp)}}{V_{DC}} = \frac{0.591\text{ V}}{15.3\text{ V}} = \mathbf{0.039}$$

The percent ripple is 3.9%.

Surge Current & the Fuse

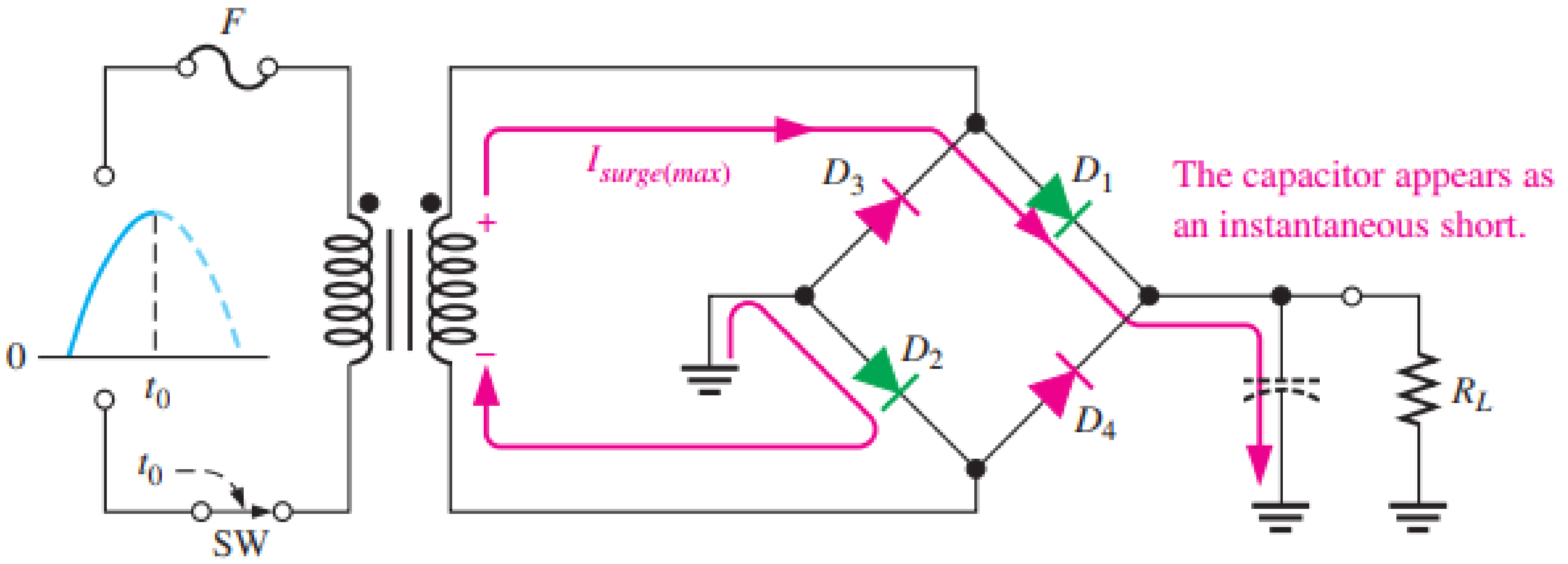


Fig 2.49 — Surge current — the uncharged capacitor looks like a short at turn-on.

Surge Current & the Fuse Cont'd

- **The Turn-On Surge**
  - Before power-on the capacitor is uncharged, so it momentarily looks like a short
  - This produces a large initial surge current through the conducting diodes
  - Worst case: the switch closes right at the secondary peak → maximum surge
- **Protecting the Supply**
  - A fuse goes in the transformer primary — a slow-blow type, to tolerate the surge
  - Primary current:

$$I_{pri} = \frac{P_{in}}{120\ \text{V}}$$

  - The fuse rating should be at least 20% larger than the calculated value of *Ipri*.

Worked Example — Fuse Rating

- **The Question**
  - A supply delivers 15 V at 0.5 A to its load from a 120 V line. What primary-side fuse rating is appropriate?
- **The Solution**
  - Output power: $P_{out} = 15\ V \times 0.5\ A = 7.5\ W$
  - In an ideal transformer, $P_{in} = P_{out}$
  - Primary current: $I_{pri} = P_{out} / V_{pri} = 7.5\ W / 120\ V = 62.5\ mA$
  - Fuse rating ≥ 20% higher: $1.2 \times 62.5\ mA \approx 75\ mA$

Quick Check #1 — Power-Supply Filters

- **Question 1**
  - In a capacitor-input filter, to what voltage does the capacitor charge?
- **Question 2**
  - What is ripple voltage, and what causes it?
- **Question 3**
  - Why is a full-wave output easier to filter than a half-wave output?
- **Question 4**
  - Write the ripple-factor formula. Does a lower or higher value mean better filtering?
- **Question 5**
  - Why is a slow-blow fuse used, and where is it placed?

Part 2 — Voltage Regulators

Why Add a Voltage Regulator?

- **The Limit of a Filter**
  - A capacitor filter reduces ripple, but some always remains — and the output still drifts with line and load changes
- **The Best Approach**
  - Combine a capacitor-input filter with a voltage regulator
  - The filter reduces the ripple into the regulator to an acceptable level (under 10%)
  - The regulator holds the output constant despite changes in input voltage, load current, or temperature
  - Together they make an excellent power supply

The Three-Terminal IC Regulator

- **Three Terminals**
  - Input, output, and a reference (or adjust) terminal
  - The regulator cuts the remaining ripple to a negligible amount
- **What's Built In**
  - An internal voltage reference, short-circuit protection, and thermal shutdown
  - Available in many fixed voltages (positive and negative) and adjustable types
  - Typically supplies one or more amps with high ripple rejection

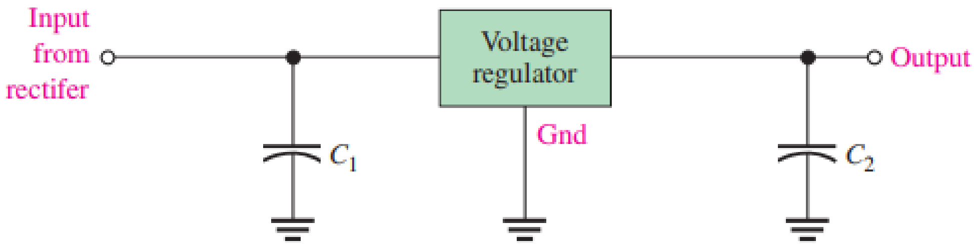


Fig 2.50 — Three-terminal IC regulator with input and output capacitors.

A Basic Regulated Supply

- **Just Two External Capacitors**
  - A fixed three-terminal regulator needs only external capacitors to work
  - A large input capacitor (the filter) sits between the input and ground
  - A small output capacitor (typically 0.1 µF to 1.0 µF) improves the transient response
- **Example**
  - A +5.0 V regulator gives a steady +5 V output for digital circuits

A Basic Regulated Supply Cont'd

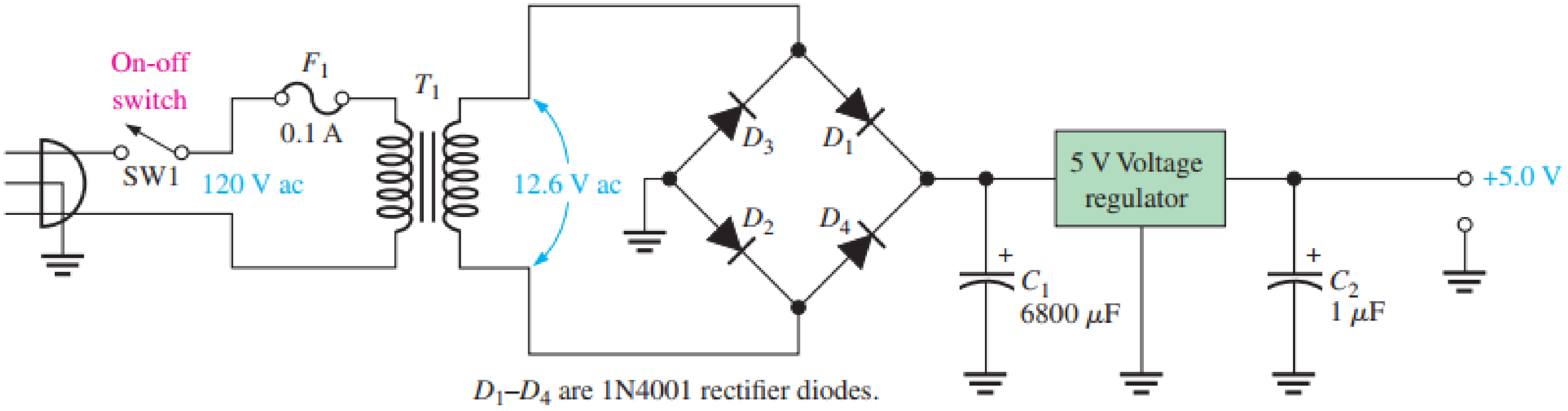


Fig 2.51 — A basic +5.0 V regulated power supply.

Line Regulation

- **Holding Steady Against Input Changes**
  - Line regulation specifies how much change occurs in the output voltage for a given change in the input voltage

$$\text{Line regulation} = \left(\frac{\Delta V_{OUT}}{\Delta V_{IN}}\right)100\%$$

- **Reading It**
  - A smaller percentage is better — the output barely moves when the line voltage varies

Worked Example — Line Regulation

- **The Question**
  - A regulator's input rises by 5 V and its output rises by 25 mV in response. What is the line regulation?
- **The Solution**
  - Line regulation = $(\Delta V_{OUT} / \Delta V_{IN}) \times 100\% = (0.025\ V / 5\ V) \times 100\% = 0.5\%$
- **The Point**
  - A 5 V swing on the input moved the output only 25 mV — that's good line regulation

Load Regulation

- **Holding Steady Against Load Changes**
  - Load regulation specifies how much change occurs in the output voltage over a certain range of load current values, usually from minimum current (no load, NL) to maximum current (full load, FL).

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right)100\%$$

- **The Terms**
  - VNL is the no-load output (minimum current); VFL is the full-load output (maximum current)
  - Again, a smaller percentage is better

Worked Example — Load Regulation

- **The Question (Example 2-9)**
  - A certain 5 V regulator has a measured no-load output voltage of 5.18 V and a full-load output of 5.15 V. What is the load regulation expressed as a percentage?
- **The Solution**

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right)100\% = \left(\frac{5.18\text{ V} - 5.15\text{ V}}{5.15\text{ V}}\right)100\% = \mathbf{0.58\%}$$

- **The Point**
  - Going from no load to full load dropped the output only 30 mV — excellent regulation

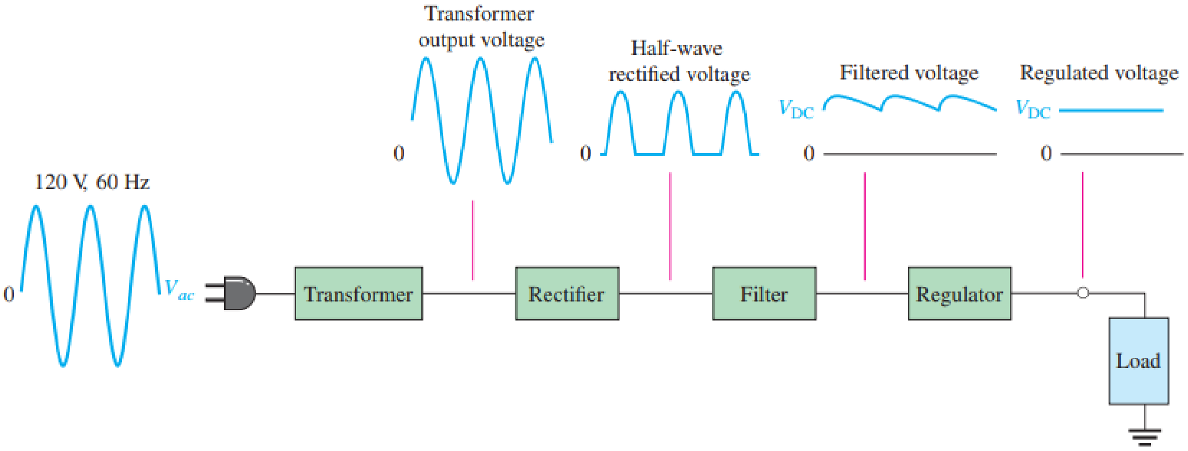
Quick Check #2 — Voltage Regulators

- **Question 1**
  - What does a voltage regulator hold constant, and against what changes?
- **Question 2**
  - Name the three terminals of an IC regulator.
- **Question 3**
  - Write the line-regulation formula.
- **Question 4**
  - Write the load-regulation formula, and define VNL and VFL.
- **Question 5**
  - For both line and load regulation, does a smaller or larger percentage indicate better performance?

Part 3 — Wrap-Up

Putting It Together — The Complete Power Supply

- **The Full Chain**
  - Transformer — steps the 120 V line down and isolates the circuit
  - Rectifier — converts AC to pulsating DC (half-wave or full-wave/bridge)
  - Filter — a capacitor smooths the pulses, leaving a small ripple
  - Regulator — holds the output constant despite line, load, and temperature changes
- **The Result**
  - A steady, low-ripple DC output ready to power electronic circuits



(a) Complete power supply with transformer, rectifier, filter, and regulator

Quick Check #3 — The Complete Supply

- **Question 1**
  - List the four stages of a complete regulated power supply, in order.
- **Question 2**
  - Which stage removes most of the ripple, and which removes what remains?
- **Question 3**
  - If you increase the filter capacitor, what happens to the ripple voltage and to the dc output?
- **Question 4**
  - Why is a filter-plus-regulator combination better than a large filter capacitor alone?

Key Takeaways from Today

- A capacitor-input filter charges to ~0.7 V below the peak, then discharges slowly through R_L — the leftover wiggle is ripple
- Full-wave (120 Hz) is easier to filter than half-wave (60 Hz): less discharge between peaks, smaller ripple
- Ripple factor $r = V_{r(pp)}/V_{DC}$ — smaller is better; lower it by increasing C or R_L
- Full-wave filter: $V_{r(pp)} \approx V_{p(rect)}/(f \cdot R_L \cdot C)$; $V_{DC} \approx [1 - 1/(2f \cdot R_L \cdot C)] \cdot V_{p(rect)}$
- Turn-on surge: an uncharged capacitor looks like a short — use a slow-blow fuse ($I_{pri} = P_{out}/V_{pri}$, rating ≥ 20% higher)
- A regulator holds the output constant vs. line, load, and temperature; filter + regulator = an excellent supply
- Line regulation = $(\Delta V_{OUT}/\Delta V_{IN}) \times 100\%$; Load regulation = $[(V_{NL} - V_{FL})/V_{FL}] \times 100\%$ — smaller percentages are better

Coming Up — Week 6: The Zener Diode & Applications

- **Next Topic — The Zener Diode (Lecture Notes pp. 75–89)**
  - A diode designed to operate in reverse breakdown at a precise voltage
  - Zener regulation, the Zener equivalent circuit, and voltage-reference applications
- **To Prepare**
  - Recall reverse breakdown from the V-I characteristic of a diode
  - Keep voltage regulation in mind — the Zener is a simple way to achieve it
- **Reminder**
  - Read pp. 75–89 before class

Thank You — Questions?